

Project Title

Full Stack Todo Management Application

1. Project Overview

This project is a **Full Stack Todo Management Application** developed using **React.js, Node.js, Express.js, and MongoDB**. The application allows users to create, view, update, and delete daily tasks (todos). It helps users organize their work efficiently by separating **pending** and **completed** tasks.

This project demonstrates practical implementation of **MERN stack concepts**, including CRUD operations, RESTful APIs, and frontend-backend integration.

2. Objectives of the Project

- To build a task management system using MERN stack
 - To perform CRUD operations on todos
 - To understand REST API integration
 - To manage application state using React
 - To follow clean project structure and best coding practices
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3. Technology Stack

Frontend

- React.js
- React Router DOM
- Axios / Fetch API
- CSS

Backend

- Node.js
- Express.js

Database

- MongoDB
- Mongoose

4. Project Architecture

The application follows a **Client–Server Architecture**:

- **Frontend (React)** handles UI rendering and user interactions
- **Backend (Node.js + Express.js)** manages API requests and business logic
- **MongoDB** stores todo data persistently

5. Core Features

- View all todos
- Create new todos
- Edit existing todos
- Delete todo

6. Functional Modules

6.1 Todo Management Module

- Add new todo tasks
- Update todo content
- Delete todos
- Mark tasks as completed
- Display pending and completed tasks

7. Project Directory Structure

Backend Structure

```
backend/
├── config/
│   └── db.js # MongoDB connection
├── models/
│   └── Todo.js # Todo schema
├── routes/
│   └── todoRoutes.js # API routes (CRUD)
├── controllers/
│   └── todoController.js # Business logic
├── server.js # Express app entry point
└── package.json
```

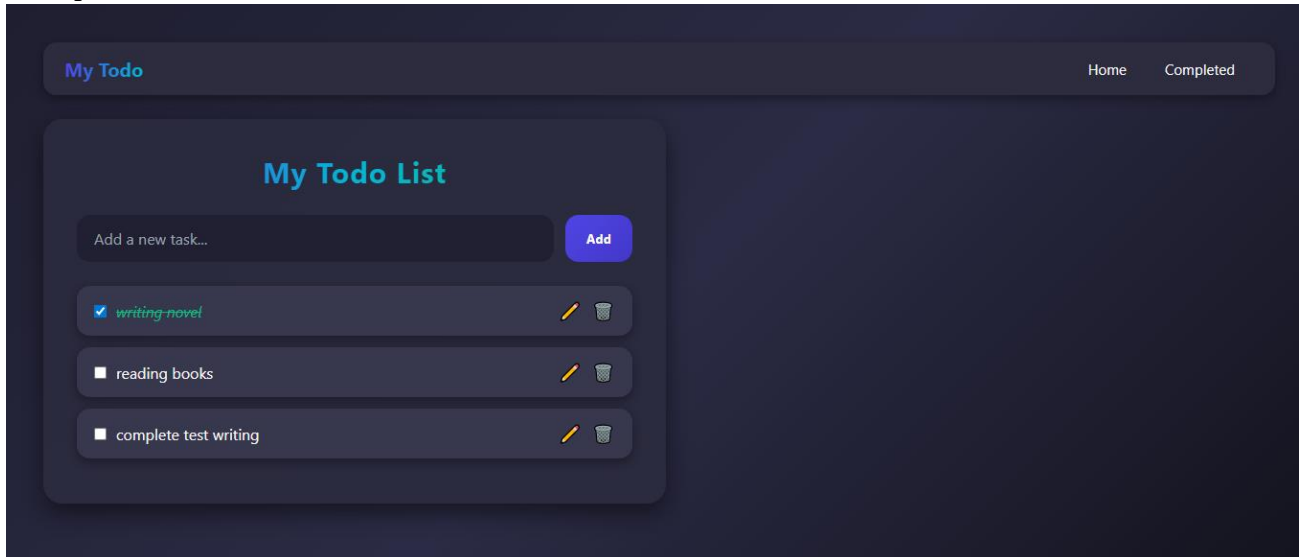
Frontend Structure

```
frontend/
├── src/
│   ├── components/
│   │   ├── TodoItem.jsx
│   │   ├── TodoInput.jsx
│   │   └── Navbar.jsx
│   ├── pages/
│   │   ├── Home.jsx
│   │   └── Completed.jsx
│   ├── api/
│   │   └── todoApi.js # Axios API calls
│   ├── App.jsx
│   └── main.jsx
└── package.json
```

8. User Interface Description

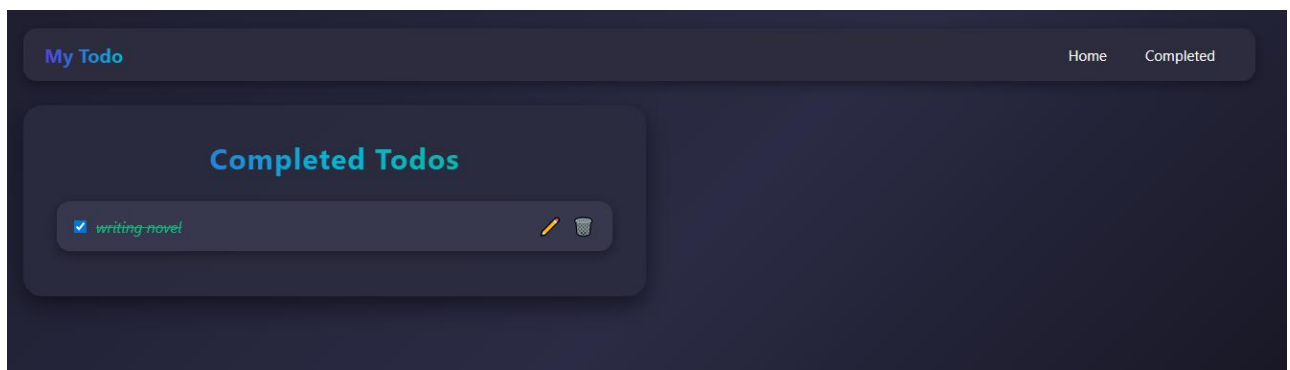
Home Page

- Displays all pending todo tasks
- Input field to add new todos
- Options to edit or delete tasks



Completed Page

- Displays completed todo tasks
- Helps users track finished work



9. Security & Validation

- Input validation on frontend
- Error handling in backend APIs
- Secure MongoDB connection using environment variables